Mini Computer Project

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NPRE 449: Nuclear Systems Engineering & Design
University of Illinois Urbana-Champaign

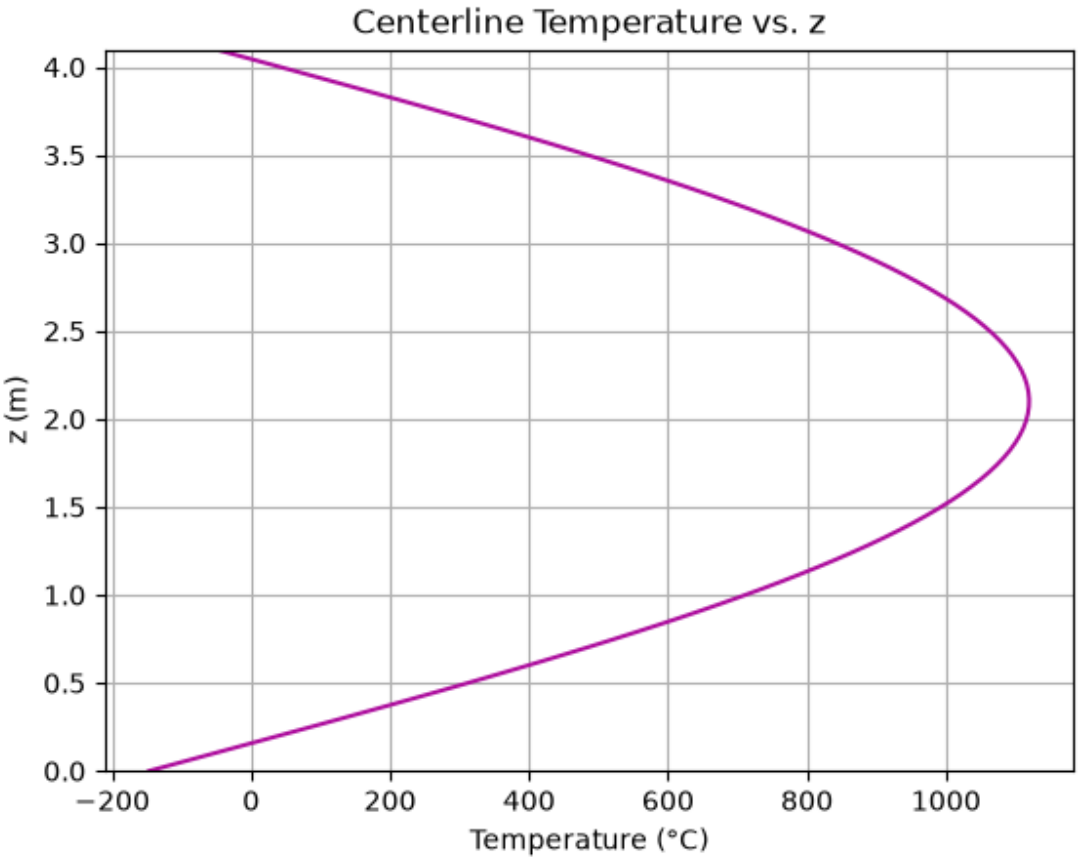
Teaching Assistant(s): Seth Stines and ~~Sohaib Malik~~
Professor: Dr. Caleb Brooks

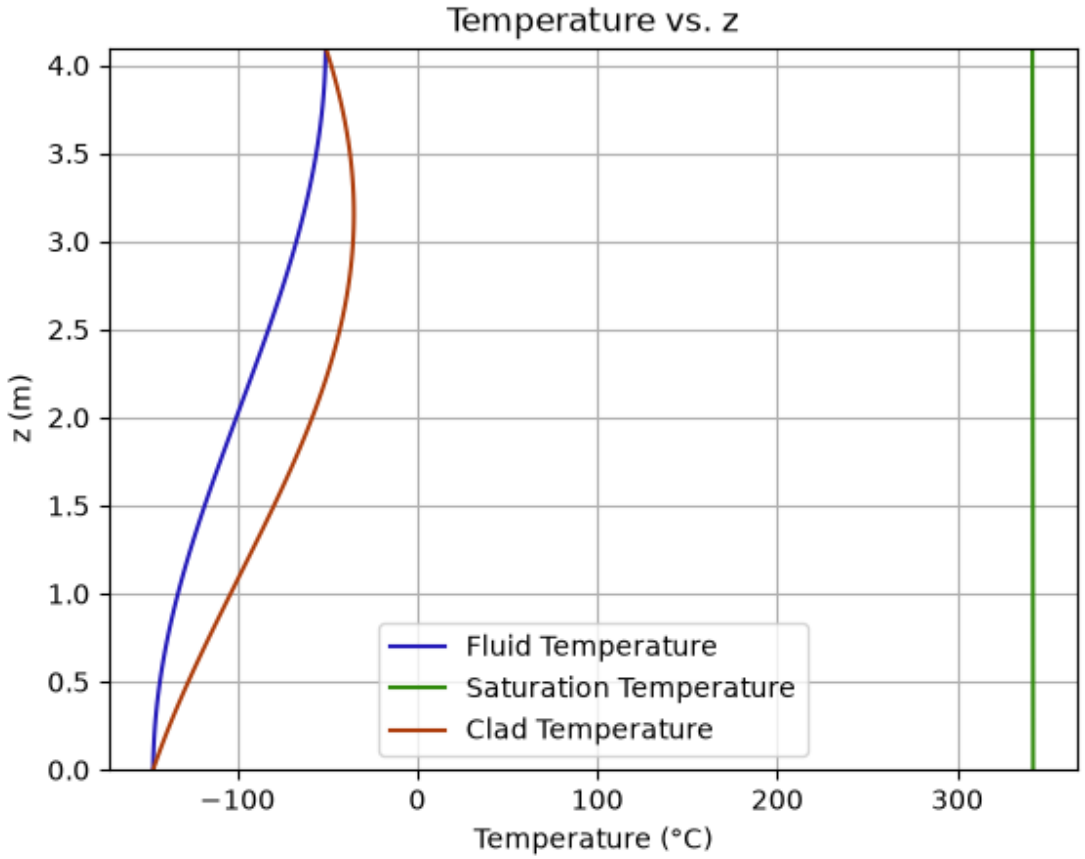
November 11, 2026

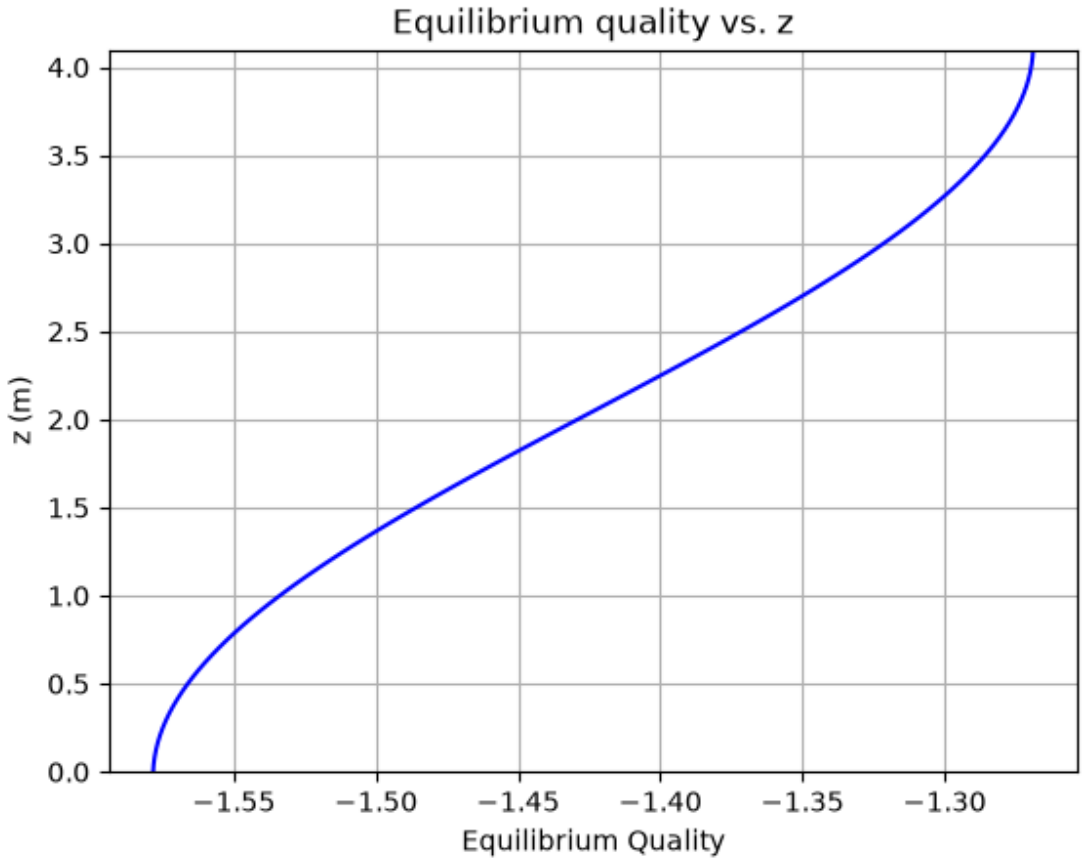
CODE HERE BELOW

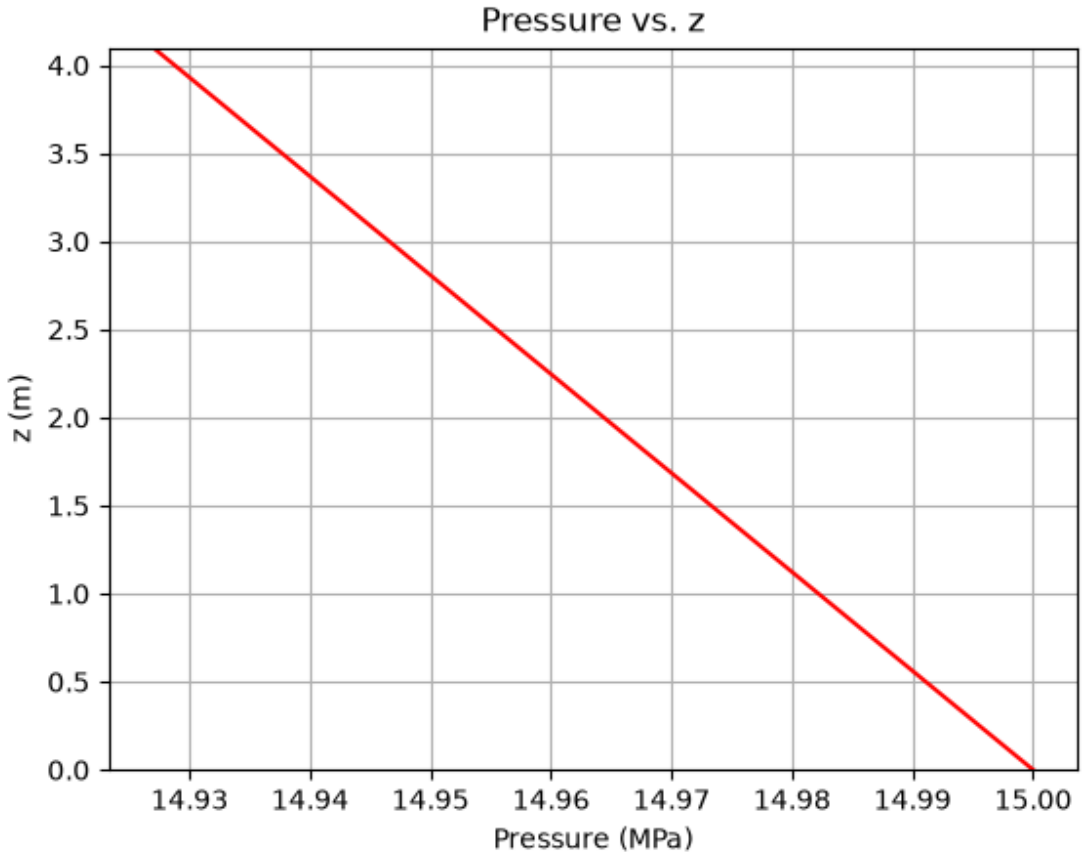
https://github.com/PeterCannon10/npre449/blob/master/simple_cp.py

1 Spatial Dependency - Q1




Temperature vs. z
z (m)
Temperature (°C)
Fluid Temperature
Saturation Temperature
Clad Temperature
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−100
0
100
200
300

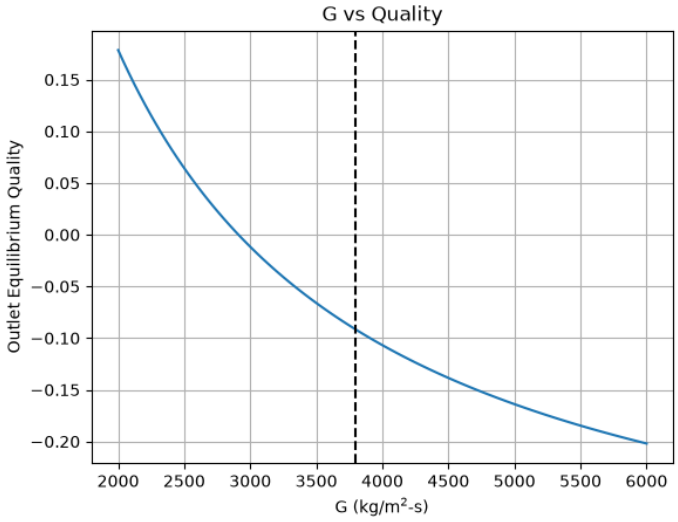

Equilibrium quality vs. z
z (m)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−1.55
−1.50
−1.45
−1.40
−1.35
−1.30
Equilibrium Quality

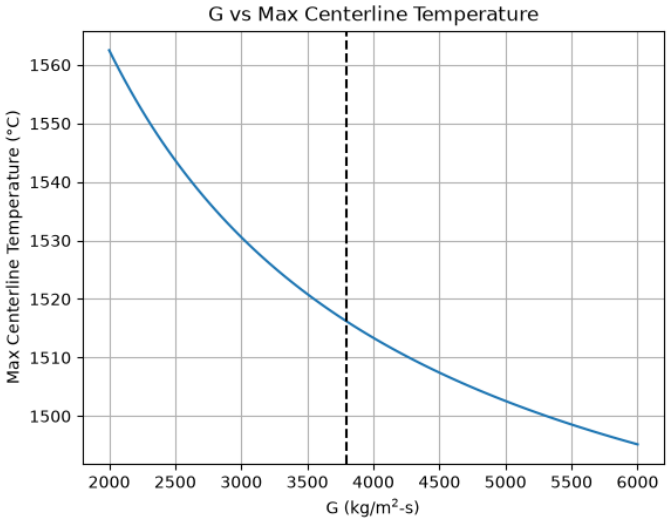

Pressure vs. z
z (m)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
14.93
14.94
14.95
14.96
14.97
14.98
14.99
15.00
Pressure (MPa)

2 Geometry & Boundary Condition Dependency - Q2

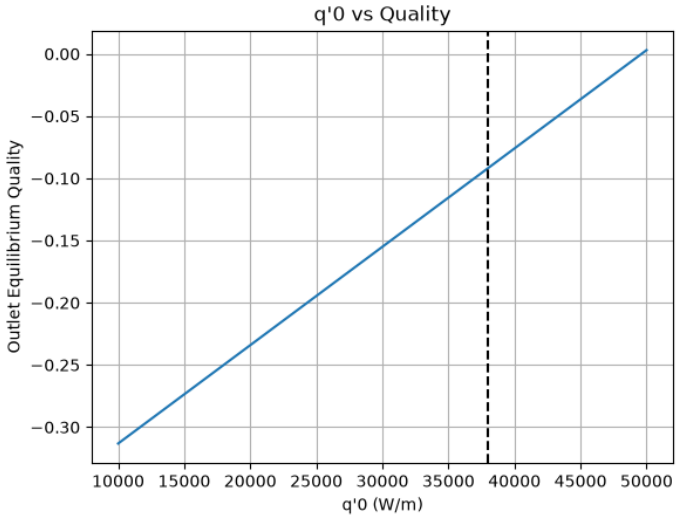
The geometry and boundary conditions included in the sensitivity study are: mass flux G, linear heat transfer rate q'_0, inlet pressure P_0, inlet fluid temperature T_0, rod diameter d_{rod}, pitch, and height h. The dotted black line corresponds to the original value on the x-axis. Each property will be varied independently of all others, treating other properties as constant. The sensitivity of each property will not be addressed directly, but can be inferred from the graph. Larger changes over short intervals for equilibrium quality and/or max centerline temperatures indicate high sensitivity and vice versa.

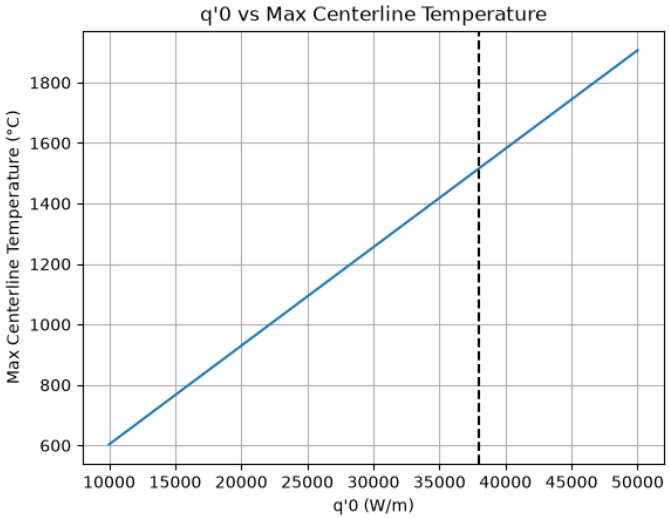
2.1 Mass Flux - G



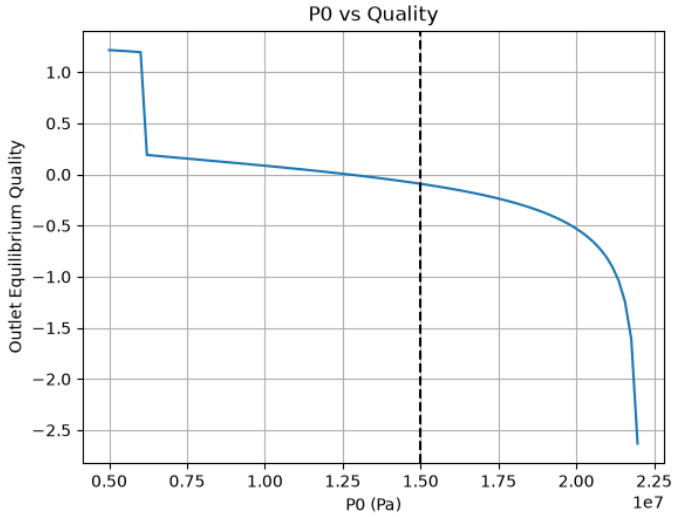


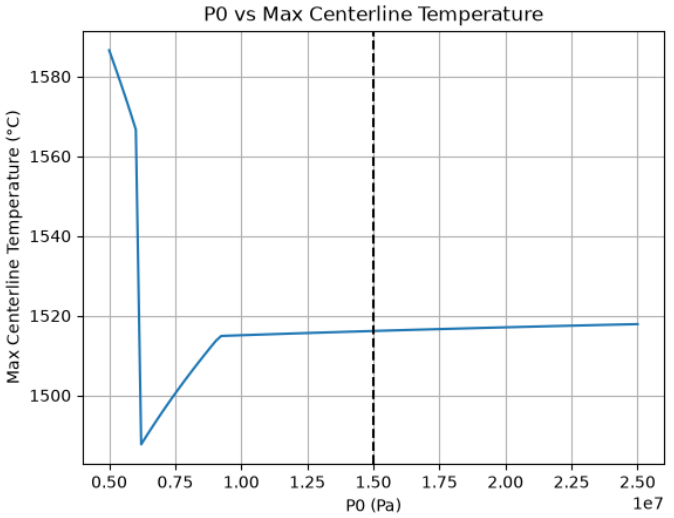
2.2 Linear Heat Transfer Rate - q_0'



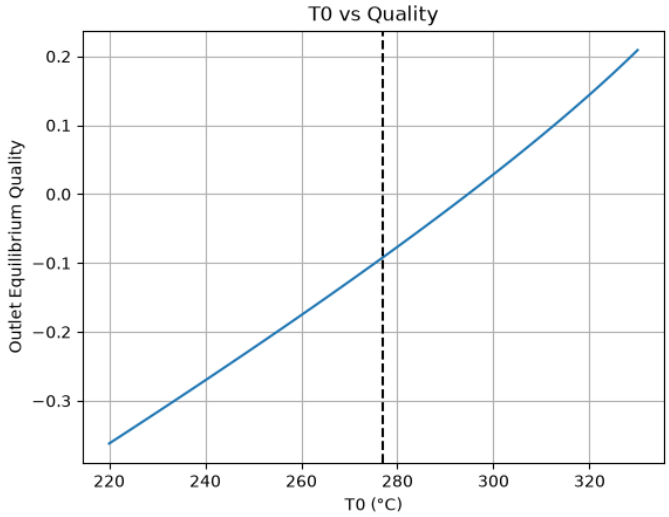


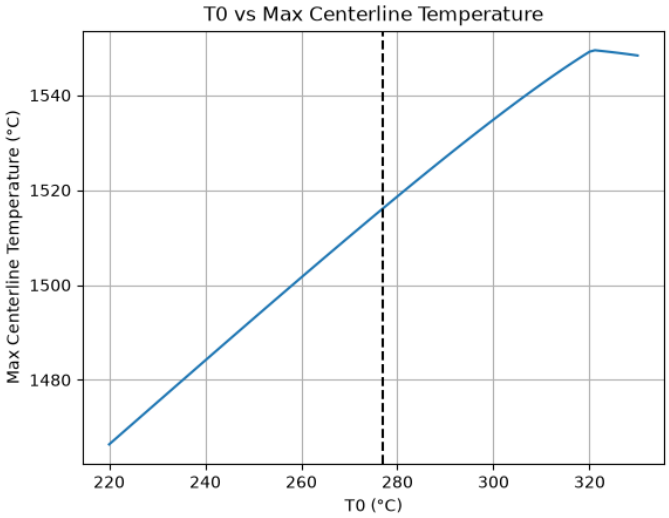
2.3 Inlet Pressure - P_0



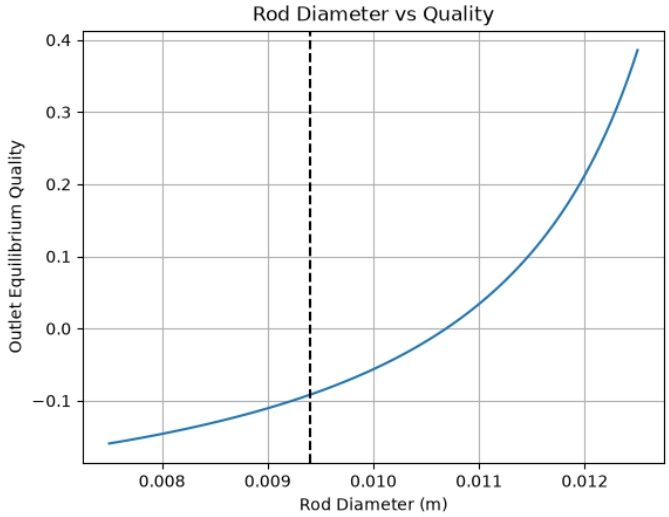


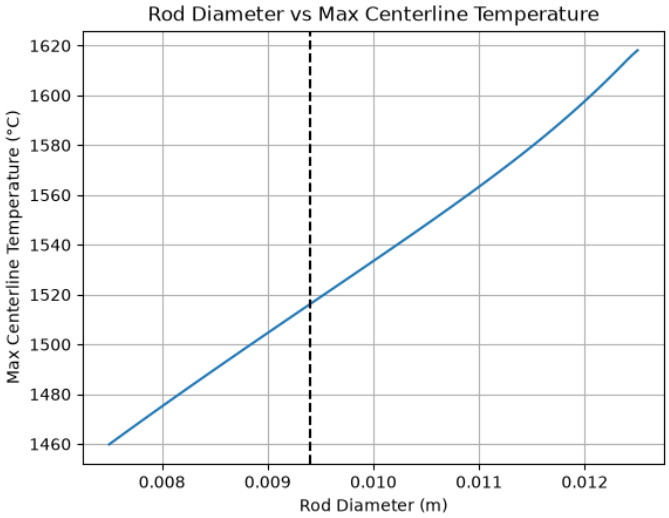
2.4 Inlet Fluid Temperature - T_0



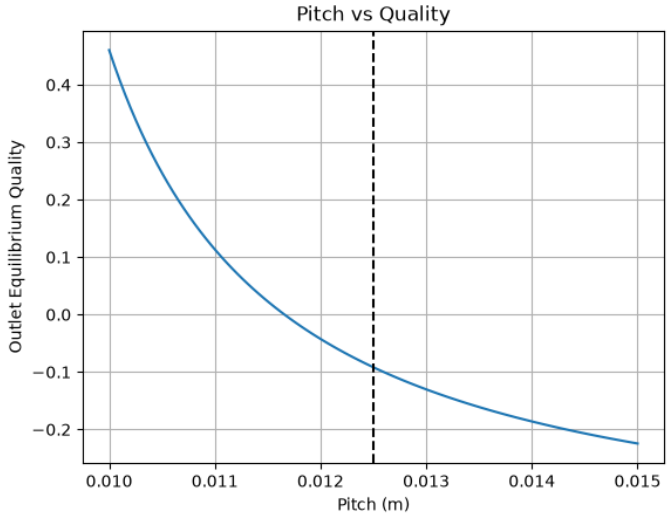


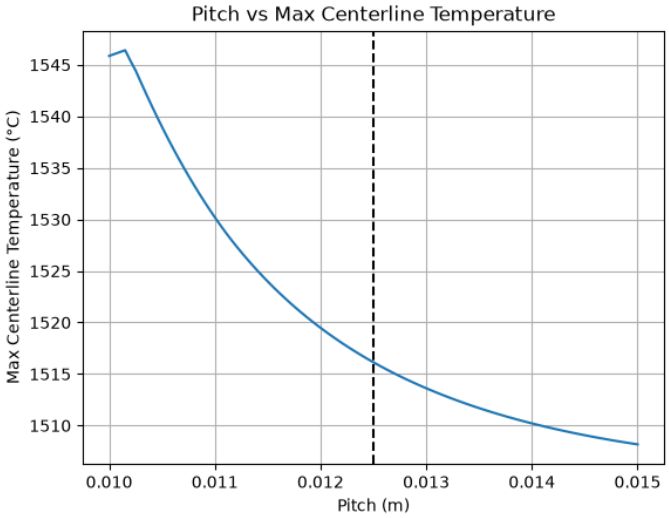
2.5 Rod Diameter - d_{rod}



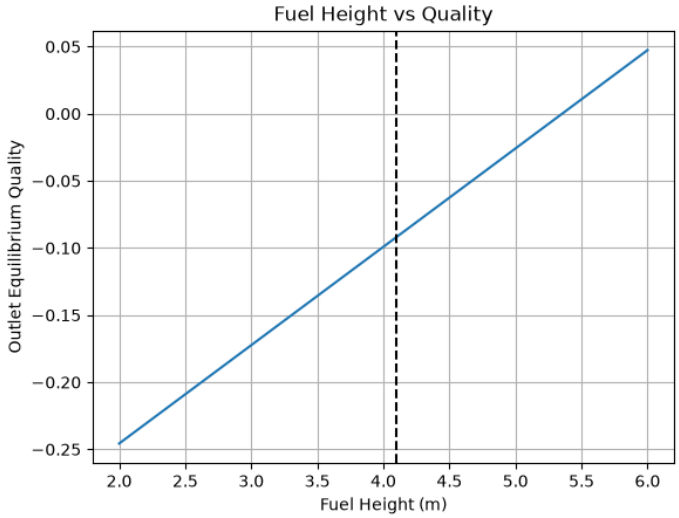


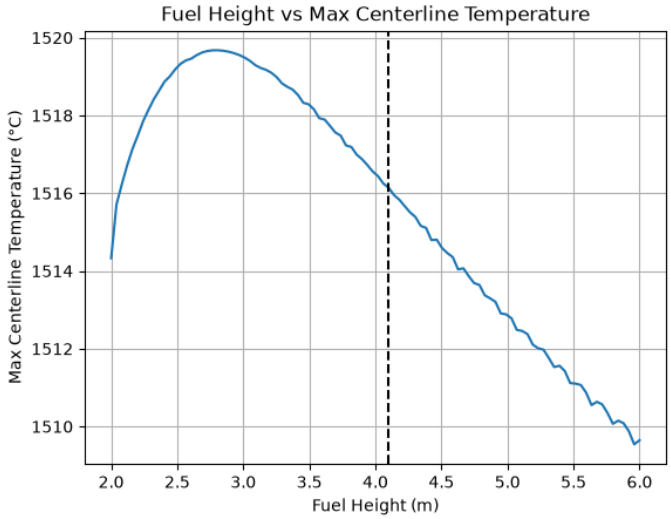
2.6 Pitch


Pitch vs Quality
Outlet Equilibrium Quality
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
0.010
0.011
0.012
0.013
0.014
0.015
Pitch (m)


Pitch vs Max Centerline Temperature
Max Centerline Temperature (°C)
1545
1540
1535
1530
1525
1520
1515
1510
0.010
0.011
0.012
0.013
0.014
0.015
Pitch (m)

2.7 Height - h


Fuel Height vs Quality
Outlet Equilibrium Quality
0.05
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
Fuel Height (m)


Fuel Height vs Max Centerline Temperature
Max Centerline Temperature (°C)
1520
1518
1516
1514
1512
1510
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
Fuel Height (m)